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Coverage optimization of wireless sensor networks based on bottle sea sheath optimization algorithm

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Abstract: To tackle the problem of inadequate placement of wireless sensor networks' sensor nodes, a proposed algorithm aims to enhance network coverage through a more optimal approach. Recognizing limitations in the Salp Swarm Algorithm (SSA) optimization, specifically its insufficient local search capability and susceptibility to local extreme values, a hybrid strategy is introduced in the search step. This strategy involves adaptive weight factor adjustments and incorporates Levy flight, achieving a better balance between global search, convergence speed, and computational accuracy. Consequently, this refinement contributes to improved effective network coverage, enhancing the overall network coverage in wireless sensor networks and facilitating faster coverage of the nodes.

1. Introduction

A WSN, or Wireless Sensor Network, is an area of detection in which multiple small, cost-efficient, and low-power sensor nodes are distributed and organized into a self-organizing, multi-hop network system. The primary objective of optimizing WSN coverage is to achieve seamless and comprehensive coverage within the specified detection area. To fulfill coverage requirements, sensor nodes are often randomly deployed, potentially creating coverage blind spots or overly dense node clusters. Blind coverage negatively impacts monitoring quality, while excessive node density can lead to channel congestion, data interference, and significant energy and cost wastage during data reception, processing, and forwarding.

In recent years, various scholars have explored the optimization of Wireless Sensor Network (WSN) node coverage by applying different algorithms, including the particle swarm algorithm, firefly algorithm, genetic algorithm, and artificial bee colony algorithm, among other options. However, these algorithms encounter challenges, including complex structures, premature convergence, low coverage rates, and reduced accuracy. Consequently, researchers have proposed several enhancement methods to address these issues. In [1], An approach using the Swarm technique for adaptive chaotic quantum particles was proposed to optimize WSN coverage, outperforming standard particle swarm, swarm of quantum particles, and chaotic quantum particle swarm. However, the achieved coverage rate of 87.15% was still deemed unsatisfactory. In [2], A refined version of the grey wolf optimization algorithm was suggested for deploying WSN nodes, yielding superior outcomes. Nevertheless, the algorithm demonstrates considerable complexity. This approach attempts to handle the distinctive structural properties of WSNs. Li and Lu [3] introduced an enhanced artificial bee colony algorithm to achieve a more uniform node distribution, yet further improvements are necessary for achieving approximate complete coverage. Fu and Hu [4] suggested using the Bayesian artificial bee colony algorithm to formulate a WSN node distribution scheme, demonstrating a higher coverage rate and increased



stability. Meanwhile, Xu et al. [5] presented a mixed strategy incorporating continuity boundary factors and dynamic weights in the ant-lion algorithm. This study concentrates on improving the effectiveness of an optimization algorithm and applying it to enhance the wireless sensor network coverage. Specifically, the research emphasizes the intelligent positioning and appropriate deployment of sensor nodes within WSNs, aiming to improve coverage rates in the designated detection area, ensuring full coverage and effective information acquisition. An enhanced MSSA (Modified Salp Swarm Algorithm, MSSA) is introduced.

2. Network coverage model

This paper explores wireless sensor network coverage using a probabilistic perception model. The model sets the probability of the target point's observation p by the node s_i $C_p(s_i, p)$ to be:

$$C_p(s_i, p) = \begin{cases} 1, & d(s_i, p) \leq R_s - r_e \\ \exp\left(\frac{-\lambda_1 \alpha_1^{\beta_1}}{\alpha_2^{\beta_2} + \lambda_2}\right), & R_s - r_e < d(s_i, p) < R_s + r_e \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

In the above equation, the variable " $d(s_i, p)$ " denotes the target's distance from the node measured in Euclidean terms. The parameter " R_s " represents the radius of the measurement range for the detector node. The sensor node's measurement reliability parameter is denoted by r_e ($0 < r_e < R_s$), $\alpha_2 = r_e + R_s - d(s_i, p)$. The measurement parameters $\alpha_1 > 0$ $\alpha_2 > 0$ and $\lambda_1 \lambda_2 \beta_1 \beta_2$ correspond to the attributes of the node. The probability threshold for detecting the target point is C_{th} and the likelihood of $C_p(s_{all}, p)$ being jointly monitored by every sensor node in the observation region is:

$$C_p(s_{all}, p) = \begin{cases} 1, & (1 - \prod_{i=1}^n [1 - C_p(s_i, p)]) \geq C_{th} \\ 0, & (1 - \prod_{i=1}^n [1 - C_p(s_i, p)]) < C_{th} \end{cases} \quad (2)$$

Here, s_{all} represents all sensor nodes combined that can monitor the target location. Considering that the region being monitored by the wireless sensor is $m \times n$ (m^2) rectangle, the monitored area is divided into $m \times n$ grids by grid partitioning, and the cells are simplified into pixel points. The proportion of covered grid numbers by Equation (3) to the total number of grids in the area under observation is the definition of WSN node coverage in this study, i.e.:

$$R_{cov} = \frac{\sum_{x_p=1}^m \sum_{y_p=1}^n C_p(s_{all}, p)}{m \times n} \quad (3)$$

3. WSN network coverage optimization algorithm

3.1 Modified bottle sea sheath swarm algorithm (MSSA)

In order to improve the network coverage, this paper optimizes the traditional sea sheath algorithm and proposes the Modified bottle sea sheath swarm algorithm (MSSA).

3.2 Algorithm parameter tuning

The leader position update formula is essential to maximize the SSA algorithm's exploration and exploitation capabilities. This parameter holds significant importance, and its formula is enhanced as follows:

$$c_1 = 0.5 * e^{(1+5l/L)^5} \quad (4)$$

3.3 Adaptive weighting factors

In the leader updating step of SSA, the issue of an endless search range is addressed by adding an adaptive weight component to the food source. A substantial inertia weight (w) is required in the algorithm's initial stage to retain effective global search capabilities. As a result, the hyperbolic tangent function is used in this work as a nonlinear method of inertia weight adjustment. The adaptive weight factor formula is expressed as follows:

$$w(t) = \frac{e^{2(1-l/L)} - e^{-2(1-l/L)}}{e^{2(1-l/L)} + e^{-2(1-l/L)}} \quad (5)$$

3.4 Levy flight perturbation strategy

A Levy flight perturbation approach is presented to get around the SSA algorithm's propensity [6] to become trapped in local optima. The algorithm can alternate at random between long and short distances thanks to this tactic. The investigation concludes that the following is a simple expression for the density function for the Levy flight step change:

$$L(s) \sim |s|^{-1-\beta}, \quad 0 < \beta \leq 2 \quad (6)$$

Then, Equation (6) refers to the Levy exercise step length. Mantegna [7] can be found at s from Equation (7):

$$s = \frac{\mu}{|\nu|^{1/\beta}} \quad (7)$$

where μ and ν are normally distributed random numbers and $\mu \sim N(0, \sigma_\mu^2)$, $\nu \sim N(0, \sigma_\nu^2)$. σ_μ and σ_ν are obtained from Equation (8):

$$\begin{cases} \sigma_\mu = \left\{ \frac{\Gamma(1+\beta) \sin(\pi\beta/2)}{\Gamma[(1+\beta)/2] \beta 2^{(\beta-1)/2}} \right\}^{1/\beta} \\ \sigma_\nu = 1 \end{cases} \quad (8)$$

Here, the parameter β takes a value in the range $0 < \beta < 2$, usually $\beta = 1.5$.

Combining the above components, the reader position is updated according to Equation (9) below:

$$x_j^i = \begin{cases} w \times F_j + c_1((ub_j - lb_j)c_2 + lb_j) \times s, & c_3 \geq 0.5 \\ w \times F_j - c_1((ub_j - lb_j)c_2 + lb_j) \times s, & c_3 < 0.5 \end{cases} \quad (9)$$

where x_j^i is the status of the first i bottled ascidian (chief) in j three-dimensional area. F_j is the location of the dietary supply in j three-dimensional area. ub_j and lb_j are the upper and lower limits of position values in j dimensional space respectively. c_2 and c_3 are uniformly distributed random numbers at $[0, 1]$.

3.5 Non-uniform gaussian variational operator

To get over the SSA algorithm's follower position update limitation—which is based only on past users and disregards information about other people—the basic follower position update formula is modified by adding a non-uniform Gaussian variation operator:

$$x_j^i = \begin{cases} \frac{1}{2}(x_j^i + x_j^{i-1}) + \Delta(l, GD_l^i), & c_4 \geq 0.5 \\ \frac{1}{2}(x_j^i + x_j^{i-1}) - \Delta(l, GD_l^i), & c_4 < 0.5 \end{cases} \quad (10)$$

$$\Delta(l, m) = m(1 - r^{(1-l/L)^b}) \quad (11)$$

$$GD_l^i = N((F_j - x_j^i), \sigma) \quad (12)$$

where $\Delta(l, GD_l^i)$ is a non-uniformly varying step, a kind of variation operator whose step is adaptively adjusted by the Gaussian distribution. GD_l^i , c_q and r are random numbers distributed uniformly in $[0, 1]$. l is the number of iterations at this time. L is the total iteration count. b is a parameter of the system that establishes non-uniformity, and its value is taken to be $[1, 5]$, and in this paper $b = 2$. F_j is the location of the dietary supply in j three-dimensional area. σ is the Gaussian distribution's standard deviation.

3.6 Incorporating cauchy gaussian variation and reverse learning strategies

Reverse learning, introduced by Tizhoosh in 2005, is a novel approach designed to identify the reverse solution corresponding to the existing resolution using a reverse learning mechanism. It assesses the comparison and stores a superior solution. To enhance individuals' ability to discover optimal solutions, the Bottle Sea Sheath algorithm incorporates the reverse learning method, represented mathematically as follows:

$$X'_{best}(t) = ub + r \oplus (lb - X_{best}(t)) \quad (13)$$

$$X_{ij}^{t+1} = X'_{best}(t) + b_1 \oplus (X_{best}(t) - X'_{best}(t)) \quad (14)$$

where $X'_{best}(t)$ is the ideal solution's opposite, ub and lb are both the upper and lower bound, respectively, r is a $1 \times d$ random number matrix following a $(0, 1)$ standard uniform distribution, and b_1 [8] is the control parameters of the information exchange.

The formula is as follows:

$$b_1 = (L - \frac{l}{L})l$$

3.7 Coverage optimization algorithm design

The MSSA algorithm-based WSN coverage optimization's goal is to maximise coverage of the observed region using a fixed number of sensor nodes R_{cov} . In this technique, the ascidian population comprises numerous individual ascidians, each of which has an equal number of dimensions and represents the coverage distribution of every sensor node. The number of dimensions of ascidian persons is twice the number of sensor nodes since the observed area is a plane with two dimensions. If the number of nodes is N , the number of individual dimensions is $2N$. Here, $2i-1$ and $2i$ represent the horizontal vertical coordinates of the node at i respectively. After the optimisation of the algorithm, the best-fit values R_{cov} and the outcome of the algorithm's optimization are the distribution coordinates of the sensor nodes that were looked for. The algorithm's individual steps are listed below.

Step 1 We enter the perceptual radius, the region edge length, the number of nodes, etc. The extent of the algorithm's population U is set to 30, and the number of iterations is 500.

Step 2 We initialize the population of bottlenose sea squirts.

Step 3 We determine each person's coverage using Equation (3) and determine the location of the individual with the largest coverage as the supply of food location *FoodPosition*.

Step 4 We calculate the adaptive weight factor w and Levy random step s , and use the equation to update the leader's Position (9).

Step 5 We figure out the non-uniform Gaussian variation operator $\Delta(l, GD_l^i)$, get the new follower position according to Equation (10), compare it with the original position, and keep it if it is better; otherwise, we discard the new position obtained.

Step 6: We mutate the global optimal position and calculate the coverage of the optimal individual after updating the position R_{cov} . If it is greater than the current *FoodPosition*, then we update *FoodPosition*.

Step 7: We determine whether the current iteration number reaches the preset iteration number. If so, we output the global optimal solution, which includes the matching node deployment's position coordinates and ideal coverage; otherwise, we jump to Step 4 for the next iteration.

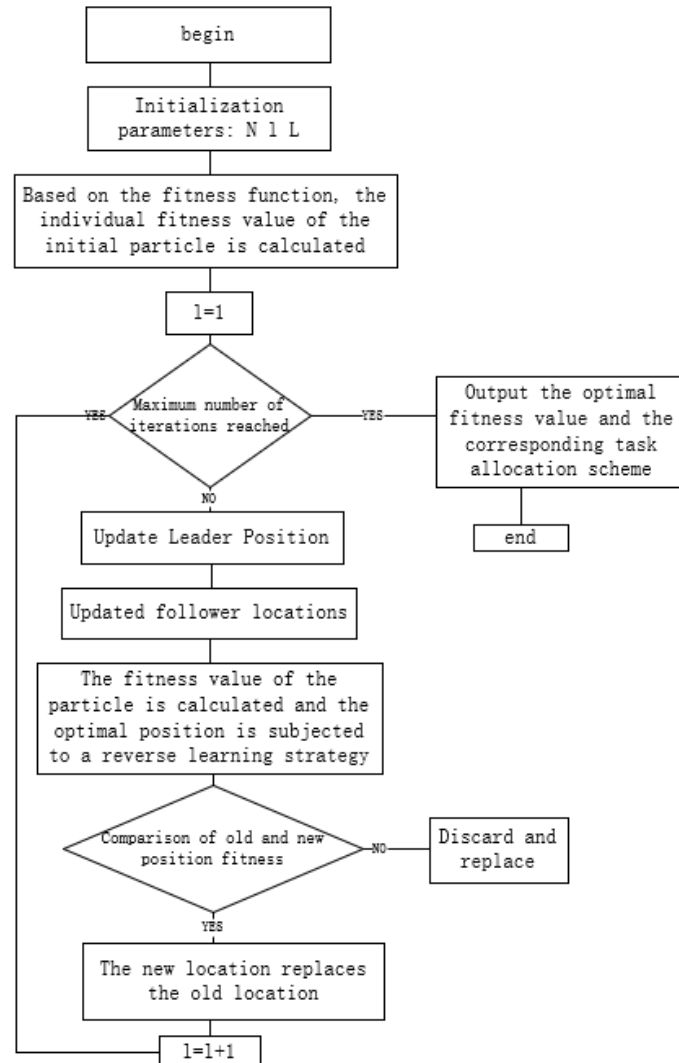


Figure 1. Flowchart for an algorithm.

Figure 1 displays the flowchart of the algorithm.

4. Experimental design and analysis of results

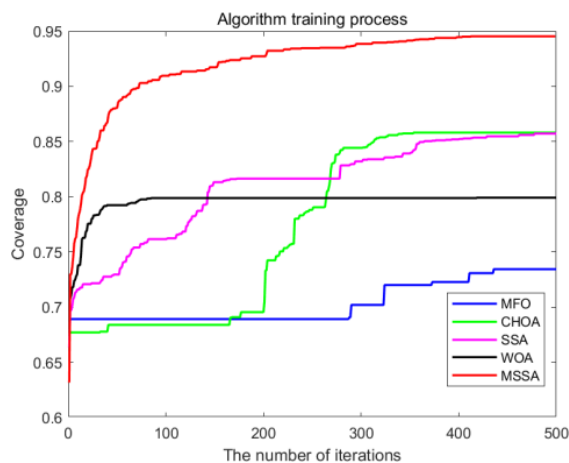
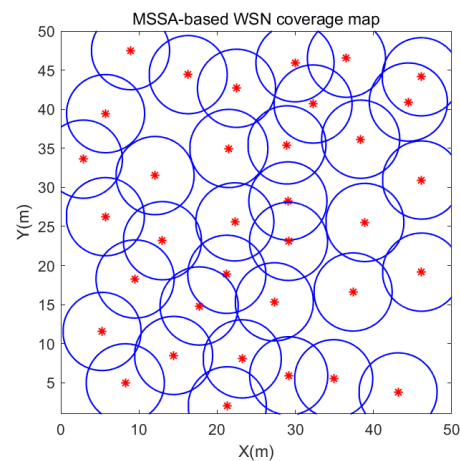
To thoroughly assess the effectiveness of the MSSA algorithm, it is juxtaposed against conventional optimization algorithms, including the Salp Swarm Algorithm (SSA), Whale Optimization Algorithm (WOA) [9], Moth-Flame Optimization (MFO) [10], and Chimp Optimization Algorithm (CHOA) [11]. The performance of the MSSA algorithm is specifically evaluated through simulation experiments, highlighting its search capability in a wireless sensor network coverage scenario. The parameter settings are outlined in Table 1, and Table 2 provides a comparative analysis of the coverage values achieved by each algorithm.

Table 1. Parameter Settings.

Parameters	Numerical value
Deployment area size/m	50*50
Number of nodes/pc	35
Node sensing radius/m	5
Node communication radius/m	10

Table 2. Comparison of Algorithm Coverage Values.

Algorithm name	Site coverage
MFO	73.8%
CHOA	83.2%
SSA	83.4%
WOA	78.5%
MSSA	94.8%

**Figure 2.** Algorithm coverage with number of iterations.**Figure 3.** MSSA coverage plot.

From Figure 2, the line graph of the coverage rate changes with the increase of iterations. We can also see that the SSA and CHOA algorithms do not easily fall into local convergence, and the algorithms have a strong ability to search for optimization in the latter stage, but the algorithms are slow to converge, and the algorithms are found to be unstable during the experimental process. The MFO algorithm and WOA algorithm have poor optimization ability for high-dimensional complex problems. The improved algorithm MSSA algorithm can find the optimization speed. It has a strong ability and does not easily fall into local convergence. It can be a good response to high-dimensional complex problems, better than the four algorithms in the comparison. The wireless sensor network coverage diagram for the upgraded MSSA algorithm is shown in Figure 3.

5. Conclusion

The coverage problem is among the main research questions for wireless sensor networks, which fundamentally reflects the ability of the whole network to perceive the monitoring area. Seeking effective algorithms to solve the wireless sensor network node deployment is a current research hotspot. Effective solution algorithms can help people deploy sensor nodes more reasonably and effectively, maximizing network coverage. This paper proposes a relatively optimal network coverage algorithm for wireless sensor particle coverage, with maximizing network coverage as the optimization objective. Through the MSSA algorithm, it realizes the improvement of the effective network coverage rate and

completes the rapid coverage of the nodes. Simulation results show that under the same experimental conditions, in comparison to the other four conventional techniques, the MSSA approach presented in this study converges more quickly and has a higher and more consistent coverage rate.

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